From traditional animal breeding to genomic selection

Genomics for Animal Sciences - Summer Workshop (Track 3)

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Animal breeding: estimating genetic merit (breeding value)

- Challenge: observed performance is influenced by both genetics and non-genetic factors
- Key idea: separate genetic effects from non-genetic effects by analyzing all available information simultaneously
    - Animal's own performance records
    - Performance of relatives
    - Pedigree (or genomic) relationships
    - Environmental factors
- Outcome: estimated breeding values that can be fairly compared across animals raised in different environments

Henderson's general linear mixed model

- *Applications of Linear Models in Animal Breeding*, by Charles R. Henderson (1984) -- bridged the gap between theoretical statistics and practical animal breeding
- Established that Best Linear Unbiased Prediction (**BLUP**) yields unbiased predictions of breeding values (BVs) for animals **with and without phenotypic records**
  - Using a numerator/additive relationship matrix ($\mathbf{A}$) to capture genetic connections between **all** individuals (key to finding BVs for unphenotyped animals)
  - Using Restricted Maximum Likelihood (REML) to estimate variance components
  - Solving mixed model equations to estimate fixed effects and BVs simultaneously
- Still being used today: **pedigree-based $\mathbf{A} \rightarrow$ genomic marker-based $\mathbf{G}$**

The general linear mixed model

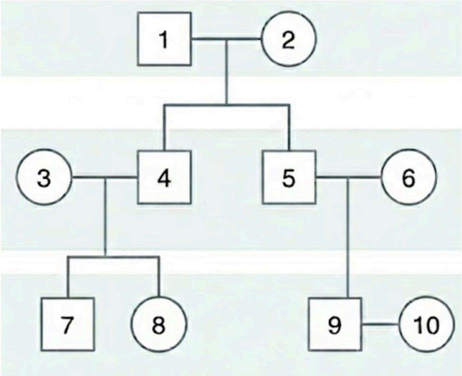
$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Zu} + \mathbf{e}$$

- $\mathbf{y}$: vector of phenotypic records
- $\boldsymbol{\beta}$: vector of fixed effects (e.g., year, season, farm, sex)
- $\mathbf{u}$: vector of breeding values (random genetic effects)
- $\mathbf{e}$: vector of random residual errors
- $\mathbf{X}, \mathbf{Z}$: design matrices relating the observations to the fixed and random effects

The variances for the random terms are:

- $\text{var}(\mathbf{u}) = \mathbf{A}\sigma_a^2$
- $\text{var}(\mathbf{e}) = \mathbf{I}\sigma_e^2$

Example of the $\mathbf{A}$ matrix



	1	2	3	4	5	6	7	8	9	10
1	1.00	0.00	0.00	0.50	0.50	0.00	0.25	0.25	0.25	0.00
2	0.00	1.00	0.00	0.50	0.50	0.00	0.25	0.25	0.25	0.00
3	0.00	0.00	1.00	0.00	0.00	0.00	0.50	0.50	0.00	0.00
4	0.50	0.50	0.00	1.00	0.50	0.00	0.50	0.50	0.25	0.00
5	0.50	0.50	0.00	0.50	1.00	0.00	0.25	0.25	0.50	0.00
6	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.50	0.00
7	0.25	0.25	0.50	0.50	0.25	0.00	1.00	0.50	0.125	0.00
8	0.25	0.25	0.50	0.50	0.25	0.00	0.50	1.00	0.125	0.00
9	0.25	0.25	0.00	0.25	0.50	0.50	0.125	0.125	1.00	0.00
10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00

The mixed model equations (MME)

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z} + \mathbf{A}^{-1}\lambda \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{y} \\ \mathbf{Z}'\mathbf{y} \end{bmatrix}$$

where λ is the ratio of the variance components: $\lambda = \sigma_e^2/\sigma_a^2$.

- Before constructing the MME, variance components need to be estimated via REML.
- BLUP for $\hat{\mathbf{u}}$ (breeding values for every animal in the analysis) can be solved: $\hat{\mathbf{u}} = (\mathbf{Z}'\mathbf{Z} + \mathbf{A}^{-1}\lambda)^{-1}\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$
  These are estimated breeding values (EBV).
- BLUP is a shrinkage estimator: the EBV for an animal with no data is shrunk toward the mean of its relatives.

Toy example of estimating BVs for four animals

- Purpose: illustrate the **flow of genetic information via $\mathbf{A}$** (without the algebraic clutter of the fixed effect adjustments)
- Simplification -- no fixed effects (no intercept)
  - valid only when **phenotypes are not raw observations, but pre-centered**
  - not a full mixed model but a random effects model
- In real-world applications, the model should always include fixed effects.

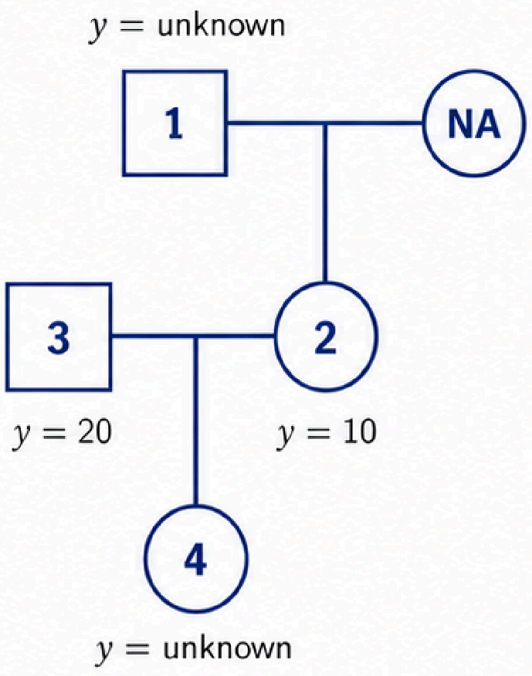
1 Data setup

We have four animals (1, 2, 3, 4):

- **Animal 1**: Founder (un-phenotyped).
- **Animal 2**: Progeny of 1 and an unknown founder (phenotyped).
- **Animal 3**: Founder (phenotyped, unrelated to 1 or 2).
- **Animal 4**: Progeny of 2 and 3 (un-phenotyped).

The Data:

- **Phenotypes** (y): Only animals 2 and 3 have records.
  - $y_2 = 10$
  - $y_3 = 20$
- **Variance Components:**
  - Additive genetic variance ($\sigma_a^2 = 20$)
  - Residual variance ($\sigma_e^2 = 40$)
  - $\lambda = \frac{\sigma_e^2}{\sigma_a^2} = \frac{40}{20} = 2$



5 Insight

1. **Direct Shrinkage**: Animals 2 and 3 have their phenotypes "shrunk" by the denominator $(1 + \lambda)$. Since $\lambda = 2$, they keep exactly $1/3$ of their performance as their genetic merit.

2. **Parental Inheritance**: Animal 1 (unphenotyped) gets exactly $5/3$, which is half of its offspring's $10/3$. This demonstrates the $1/2$ genetic flow from parent to child.

3. **Progeny Prediction:** Animal 4 (unphenotyped) gets exactly 5, which is the mid-parent average: $(10/3 + 20/3)/2 = 30/6 = 5$.

2 Forming vector and matrices

$$y = \begin{bmatrix} 10 \\ 20 \end{bmatrix} \qquad Z = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1/2 & 0 & 1/4 \\ 1/2 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1/2 \\ 1/4 & 1/2 & 1/2 & 1 \end{bmatrix} \qquad A^{-1} = \begin{bmatrix} 4/3 & -2/3 & 0 & 0 \\ -2/3 & 11/6 & 1/2 & -1 \\ 0 & 1/2 & 3/2 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}$$

3 Solve MME

$$\begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + A^{-1}\lambda \end{bmatrix} \begin{bmatrix} \hat{\beta} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

where $\lambda = 2$ and A is the numerator relationship matrix.

4 Final predicted BVs

$$\hat{u} = \begin{bmatrix} 5/3 \\ 10/3 \\ 20/3 \\ 5 \end{bmatrix} \begin{matrix} \text{Animal 1} \\ \text{Animal 2} \\ \text{Animal 3} \\ \text{Animal 4} \end{matrix}$$

R code for the toy example

```r
# --- 0. Clear Environment ---
rm(list = ls())

# --- 1. Pedigree Matrix (A) - 4x4 ---
# Rows represent animals 1, 2, 3, 4.
A <- matrix(c(1.0,  0.5,  0.0,  0.25,
              0.5,  1.0,  0.0,  0.5,
              0.0,  0.0,  1.0,  0.5,
              0.25, 0.5,  0.5,  1.0),
            nrow = 4, byrow = TRUE)

# --- 2. Design Matrix (Z) - 2x4 ---
# Explicitly defines 2 rows (phenotypes) and 4 columns (animals).
# Animal 2 gets Record 1; Animal 3 gets Record 2.
Z <- matrix(c(0, 1, 0, 0,  # Row 1: Phenotype for Animal 2
              0, 0, 1, 0), # Row 2: Phenotype for Animal 3
            nrow = 2, byrow = TRUE)

# --- 3. Phenotypes (y) and k ---
y <- matrix(c(10, 20), nrow = 2) # 2x1 Column Matrix
k <- 2 # k = lambda in the diagram

# --- 4. The MME Setup ---
# Henderson's Mixed Model Equations: (Z'Z + A^-1 * k) * u = Z'y
A_inv <- solve(A)
ZZ    <- t(Z) %*% Z         # Result: 4x4
Zy    <- t(Z) %*% y         # Result: 4x1

# --- 5. Solving for Breeding Values (u_hat) ---
LHS   <- ZZ + (A_inv * k)
u_hat <- solve(LHS, Zy)

# --- 6. Results ---
cat("--- MME Solutions (u_hat) ---\n")
print(as.vector(u_hat))
```

With genomic data

- We use $\mathbf{G}$ (genomic relationship matrix) instead of the traditional pedigree $\mathbf{A}$ matrix
  - BLUP $\Rightarrow$ GBLUP, and EBV $\Rightarrow$ GEBV
- Comparison
  - $\mathbf{A}$ (expectation): e.g., full siblings share exactly 50% of their alleles.
  - $\mathbf{G}$ (realization): captures the actual Mendelian sampling. Some full-sibs might share 42% and others 58%.

How to build $\mathbf{G}$?

$$\text{Raw genotype} = \begin{bmatrix} 1 & 0 & -1 & 1 & 0 \\ -1 & 1 & 0 & 0 & -1 \\ 0 & -1 & 1 & -1 & 1 \end{bmatrix}$$

(three animals on five SNP markers; coded by aa=-1, Aa=0, AA=1)

Let $\mathbf{M}$ be the **column-centered** matrix of the above raw genotypes,

$$\mathbf{G} = \mathbf{M}\mathbf{M}' / \sum_{i=1}^{5} 2p_i q_i$$

$\sum_{i=1}^{5} 2p_i q_i$ is the sum of the variances of all markers. This scales $\mathbf{G}$ so that the average diagonal element is approximately 1, making it comparable to the pedigree $\mathbf{A}$ matrix. (VanRaden, 2008, J. Dairy Sci.)

The `rrBLUP` R package

- User manual: https://jendelman.r-universe.dev/rrBLUP/doc/manual.html
- Implements two **equivalent** models in a single function `mixed.solve()`, depending on your input parameter:
    i. marker perspective ("RR-BLUP" or "SNP-BLUP")
        - You provide **SNP genotype matrix** -- model estimates effect of each SNP. You sum them up to get the animal's EBV.
    ii. animal perspective ("G-BLUP")
        - You provide **genomic relationship matrix** -- model estimates EBV of each animal.

Code demo using `rrBLUP`

Access the "rrBLUP" code

- Simulated 100 animals (split into training set and validation set), 500 markers, i.e., the "small *n*, large *p*" problem
- Simulated a heritability of 0.45
- Under "small *n*, large *p*" and low/moderate heritability, the model tends to fit the noise, i.e., **overfitting**
  - Artificially inflated training accuracy
  - Low and highly fluctuating validation accuracy
  - Inaccurate and fluctuating heritability estimate
  - **Solution**: feeding the model with more phenotypes

```
--- Heritability validation ---
TRUE Simulated h2 : 0.450000
Estimated G-BLUP  : 0.247283
Estimated RR-BLUP : 0.247283

--- Prediction equivalence on training animals (first 20) ---
      Animal True_BV   GBLUP_EBV  RRBLUP_EBV
1   Animal_1     4.4  1.93574248  1.93573998
2   Animal_2    14.5 10.60368019 10.60366782
3   Animal_3    10.7  0.98872971  0.98872784
4   Animal_4     1.4  3.62345449  3.62345105
5   Animal_5    -0.4  3.42570395  3.42569974
6   Animal_6    -4.7  1.09661757  1.09661643
7   Animal_7     8.2  7.38760250  7.38759385
8   Animal_8    -9.4 -7.32802126 -7.32801350
9   Animal_9     3.6  1.00603188  1.00603021
10 Animal_10    -4.6 -3.56343140 -3.56342673
11 Animal_11    -7.1 -2.20900317 -2.20900077
12 Animal_12    -4.0 -3.90671687 -3.90671340
13 Animal_13    -7.6  0.14828603  0.14828502
14 Animal_14    -0.1  0.51663498  0.51663426
15 Animal_15    -1.6 -0.06444747 -0.06444832
16 Animal_16   -10.1 -3.21654525 -3.21654110
17 Animal_17   -11.7 -3.12613131 -3.12612765
18 Animal_18    -5.8 -0.28498575 -0.28498545
19 Animal_19     4.7  0.84124039  0.84123981
20 Animal_20     0.5 -3.26210947 -3.26210465

--- Prediction equivalence on validation animals ---
       Animal True_BV   GBLUP_EBV  RRBLUP_EBV
1   Animal_81    -4.2  0.58370189  0.58370180
2   Animal_82    -1.6 -0.61101812 -0.61101727
3   Animal_83    -1.5 -0.07728269 -0.07728259
4   Animal_84    -1.2 -1.87344517 -1.87344275
5   Animal_85   -24.2 -1.22187943 -1.22187773
6   Animal_86     3.8 -0.18160803 -0.18160804
7   Animal_87    -0.4  1.45981267  1.45981066
8   Animal_88   -13.0  0.36946894  0.36946849
9   Animal_89     5.1 -0.45512092 -0.45512015
10  Animal_90    -6.2 -1.15924397 -1.15924259
11  Animal_91     5.9 -2.31187459 -2.31187151
12  Animal_92    -6.1 -0.58482067 -0.58482021
13  Animal_93    -6.1  0.11050862  0.11050874
14  Animal_94     9.9  0.13077819  0.13077784
15  Animal_95    17.4  3.53632135  3.53631723
16  Animal_96    -5.7 -0.27242672 -0.27242653
17  Animal_97    -4.9 -1.31334300 -1.31334160
18  Animal_98    -4.6  0.39689202  0.39689144
19  Animal_99     1.5 -1.87190988 -1.87190764
20 Animal_100    -7.2  1.79778300  1.79778086

--- Accuracy (correlation b/w TBV and EBV) ---
Training data accuracy: 0.742140
Validation data accuracy: 0.284866
```

Discrepancy b/w true h2 and estimated h2, due to overfitting

GBLUP and RR-BLUP produced the same predictions on both training (with phenotypes) and validation (no phenotypes) sets

High training accuracy and low validation accuracy, due to overfitting. Also, with only 20 animals for validation, the result fluctuates in each run.

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AG2Pi
AGRICULTURAL GENOME TO PHENOME INITIATIVE

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